

Question 11 – Stage 1

The line segment with end points $(1, 2)$ and $(12, a)$ (where $a > 2$) is rotated through 360° about the x -axis to create a frustum of a cone.

The same line segment is rotated through 360° about the y -axis to create another frustum of a cone.

If the difference in volumes of these two frustums is 50 cubic units, find the sum (to 3d.p.) of the possible values of a .

Working

around X

$$\begin{aligned} V_1 &= \frac{\pi h}{3}(r_m^2 + r_m r_M + r_M^2) \\ &= \frac{11\pi}{3}(2^2 + 2a + a^2) \\ &= \frac{11\pi}{3}(4 + 2a + a^2) \\ &= \frac{44\pi}{3} + \frac{22\pi}{3}a + \frac{11\pi}{3}a^2 \end{aligned}$$

around Y

$$\begin{aligned} V_2 &= \frac{\pi h}{3}(r_m^2 + r_m r_M + r_M^2) \\ &= \frac{\pi(a-2)}{3}(1^2 + 12 + 12^2) \\ &= \frac{157\pi}{3}(a-2) \\ &= \frac{157\pi}{3}a - \frac{314\pi}{3} \end{aligned}$$

Solving

$$|V_2 - V_1| = 50$$

V1 Greater

$$\begin{aligned} \frac{157\pi}{3}a - \frac{314\pi}{3} &= \frac{44\pi}{3} + \frac{22\pi}{3}a + \frac{11\pi}{3}a^2 - 50 \\ 0 &= \left(\frac{358\pi}{3} - 50\right) - 45\pi a + \frac{11\pi}{3}a^2 \\ a &= 9.210464\dots, 3.062263\dots \end{aligned}$$

V2 Greater

$$\begin{aligned} \frac{157\pi}{3}a - \frac{314\pi}{3} &= \frac{44\pi}{3} + \frac{22\pi}{3}a + \frac{11\pi}{3}a^2 + 50 \\ 0 &= \left(\frac{358\pi}{3} + 50\right) - 45\pi a + \frac{11\pi}{3}a^2 \\ a &= 7.013241\dots, 5.259486\dots \end{aligned}$$

Sum

$$\begin{aligned} \sum a &= 9.210464 + 3.062263 + 7.013241 + 5.259486 \\ &= 24.545454\dots \\ &\approx 24.545 \end{aligned}$$